

BEE 4750 Homework 2: Systems Modeling and Simulation

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Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```

Activating project at `c:\Users\mohog\hw2-moho`
Installed JpegTurbo_jll          v3.1.3+0
Installed InvertedIndices        v1.3.1
Installed Accessors              v0.1.42
Installed WorkerUtilities        v1.6.1
Installed SentinelArrays         v1.4.8
Installed InlineStrings          v1.4.5
Installed Roots                  v2.2.10
Installed DataFrames             v1.8.0
Installed WeakRefStrings         v1.4.2
Installed PooledArrays           v1.4.3
Installed TableTraits            v1.0.1
Installed DataValueInterfaces    v1.0.0
Installed Ghostscript_jll        v9.55.1+0
Installed ConstructionBase       v1.6.0
Installed PrettyTables           v3.0.8
Installed CompositionsBase       v0.1.2
Installed Crayons                v4.1.1
Installed InverseFunctions        v0.1.17
Installed CommonSolve            v0.2.4
Installed Tables                 v1.12.1
Installed Glib_jll               v2.86.0+0
Installed Latexify                v0.16.10
Installed IteratorInterfaceExtensions v1.0.0
Installed FilePathsBase          v0.9.24
Installed StringManipulation     v0.4.1
Installed CSV                    v0.10.15
Precompiling project...
1691.4 ms  IteratorInterfaceExtensions
1125.5 ms  CommonSolve
1115.4 ms  DataValueInterfaces
1150.6 ms  WorkerUtilities
1213.4 ms  InverseFunctions
 946.3 ms  CompositionsBase
 984.2 ms  InvertedIndices
1185.6 ms  ConstructionBase
1693.3 ms  InlineStrings
1346.6 ms  PooledArrays
3048.8 ms  Crayons
1958.1 ms  JpegTurbo_jll
1130.0 ms  TableTraits
3766.4 ms  SentinelArrays
2516.7 ms  FilePathsBase

```

2762.8 ms	Glib_jll
1176.8 ms	LogExpFunctions → LogExpFunctionsInverseFunctionsExt
1262.0 ms	InverseFunctions → InverseFunctionsDatesExt
2336.7 ms	InverseFunctions → InverseFunctionsTestExt
5026.2 ms	StringManipulation
1130.0 ms	ConstructionBase → ConstructionBaseLinearAlgebraExt
1217.1 ms	CompositionsBase → CompositionsBaseInverseFunctionsExt
2172.0 ms	Unitful → InverseFunctionsUnitfulExt
1380.5 ms	InlineStrings → ParsersExt
1962.5 ms	Unitful → ConstructionBaseUnitfulExt
2004.4 ms	Libtiff_jll
1165.8 ms	FilePathsBase → FilePathsBaseMmapExt
1969.5 ms	Tables
3095.1 ms	Ghostscript_jll
3321.3 ms	FilePathsBase → FilePathsBaseTestExt
2780.0 ms	Cairo_jll
2624.0 ms	WeakRefStrings
3956.6 ms	Qt6Base_jll
2681.9 ms	HarfBuzz_jll
6125.7 ms	Accessors
2399.0 ms	libass_jll
9145.8 ms	Latexify
2899.9 ms	Pango_jll
5894.5 ms	Qt6ShaderTools_jll
1993.0 ms	Accessors → LinearAlgebraExt
2143.1 ms	Accessors → TestExt
2340.8 ms	Accessors → UnitfulExt
2716.2 ms	Latexify → SparseArraysExt
3934.4 ms	FFMPEG_jll
4236.7 ms	UnitfulLatexify
3128.5 ms	FFMPEG
3804.6 ms	GR_jll
8480.5 ms	Qt6Declarative_jll
9855.2 ms	Roots
2407.7 ms	Roots → RootsUnitfulExt
10339.5 ms	GR
37852.8 ms	CSV
61530.2 ms	PrettyTables
78861.6 ms	DataFrames
6634.0 ms	Latexify → DataFramesExt
122084.5 ms	Plots
7052.1 ms	Plots → UnitfulExt

57 dependencies successfully precompiled in 180 seconds. 162 already precompiled.

```
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO₂ concentrations, and ocean CO₂ concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

System components * Global mean temperature * Atmospheric CO₂ concentrations * Ocean CO₂ concentrations

Interconnections * Atmospheric CO₂ concentrations INCREASES global mean temperature * Ocean CO₂ concentrations DECREASES atmospheric CO₂ concentrations * Global mean temperature DECREASES Ocean CO₂ concentrations

The above relationships, as described in Woodard et al. (2019), reflect how changes in Earth's carbon balance relate to mean temperature. Increasing the atmospheric CO₂ concentrations will trap more carbon which stores more heat, thus increasing temperature. In contrast, the Earth's oceans act as a carbon sink and thus when ocean CO₂ concentrations increase, the amount of CO₂ in the immediate atmosphere is expected to decrease. However, an increase in global mean temperature, including an increase in ocean temperatures, can decrease the natural ability of oceans to dissolve CO₂ due to increased stratification and decreased solubility in accordance with Henry's Law.

```
Pkg.add("GraphRecipes")
Pkg.add("Plots")
using GraphRecipes
using Plots
```

```
Resolving package versions...
No Changes to `C:\Users\mohog\hw2-moho\Project.toml`
No Changes to `C:\Users\mohog\hw2-moho\Manifest.toml`
Resolving package versions...
No Changes to `C:\Users\mohog\hw2-moho\Project.toml`
No Changes to `C:\Users\mohog\hw2-moho\Manifest.toml`
```

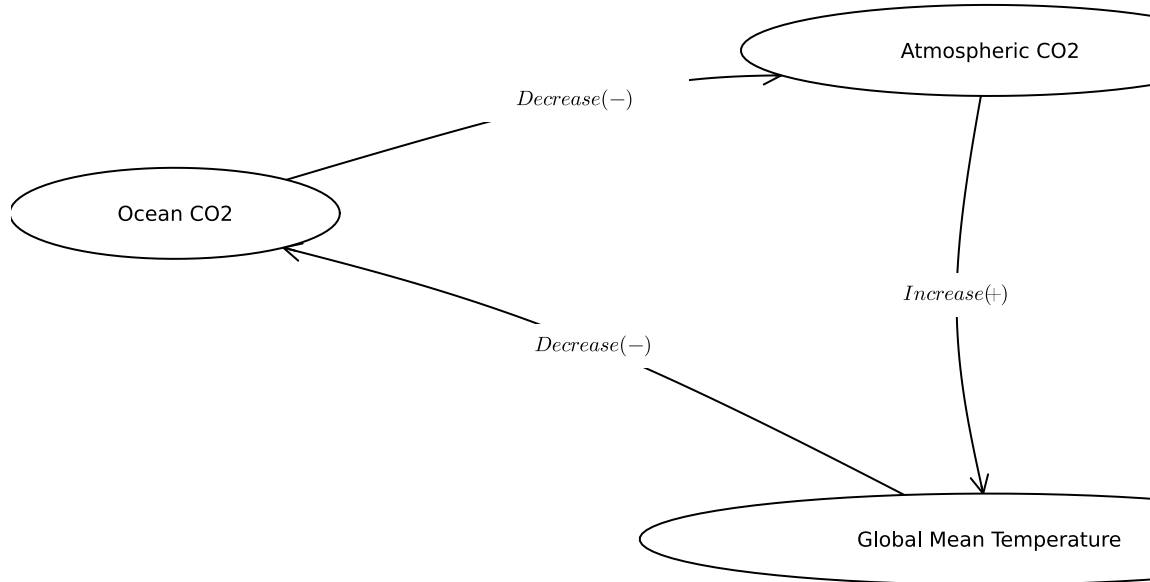
```

A = [0 0 1;
     1 0 0;
     0 1 0]

names = ["Atmospheric CO2", "Ocean CO2", "Global Mean Temperature"]
edge_labels = Dict((2, 1) => L"Decrease (-)", (1,3) => L"Increase (+)", (3, 2) => L"Decrease (-)")
shapes=[:ellipse :ellipse :ellipse]
xpos = [-0.25, -1.5, -0.25]
ypos = [0.75, 0.5, 0]

p = graphplot(A, names=names, edgelabel=edge_labels, markersize=0.12, markershapes=shapes, margin=10)
display(p)

```



Considering the overall feedback between temperature and the ocean carbon cycle, this graph suggests that increases in global mean temperature due to increases in atmospheric carbon dioxide levels from human activities such as burning fossil fuels can disrupt the extent of the ocean's natural carbon cycle by reducing its carbon uptake capabilities.

Tip

Think about Henry's law for CO₂.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/out-flows between these boxes. How did you decide how many boxes were needed?

Control volume * River: 10 km/day

Inflow * 0.36/day

Outflow * 54 kg/km/day

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

Tip

Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Problem 2.3

Determine if the system is in compliance with a regulatory limit of $2.3 \text{ kg}/(1000\text{m}^3)$. You can do this analytically or computationally.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized “Snowball Earth”. In this problem, we’ll introduce a simple model of the Earth’s energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C \frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A-BT)}_{\text{outgoing radiation}} + \underbrace{a \ln \left(\frac{[CO_2]}{[CO_2]_{PI}} \right)}_{\text{greenhouse effect}}, \quad (1)$$

where:

- T is the Earth’s global mean temperature (in °C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be $51 \text{ J/m}^2/\text{°C}$;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproterozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3 \text{ W/m}^2/\text{°C}$ (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2 \text{ W/m}^2$ (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1 \text{ yr}$.

```
# this function computes the increment by which the concentration
# is updated at each step
function box_simulate_timestep(T, x, S, C, A, B)
    return ((1-x) * S)/(4*C) - (A-(B*T))/C
end
```

```

# this function loops over the timesteps to simulate
# the concentration series
function climate_simulate(x, S, C, A, B, T, Tf, Δt)
    # initialize T storage
    # for code simplicity we make the array length T+1 so
    # index 1 is T
    steps = Int64(Tf / Δt)
    Ts = zeros(steps + 1)
    Ts[1] = T
    for t = 1:steps
        Ts[t+1] = Ts[t] +
            Δt * box_simulate_timestep(Ts[t], x, S, C, A, B)
    end
    # the first element of T is the initial condition
    return Ts
end
Δt = 1.0 #Time step of 1 year
Tf = 10 #End-time
T = 14 #Pre-industrial temperature
C = 51 #J/m^2/degrees C
x = 0.3 #Planetary albedo alpha
S = 1272 #W/m^2 #Neoproterozoic value
A = 221.2 #W/m^2
B = -1.3 #W/m^2/degrees C
Ts = climate_simulate(x, S, C, A, B, T, Tf, Δt)

```

11-element Vector{Float64}:

```

14.0
13.670588235294119
13.349573241061131
13.03674098197526
12.731882878513145
12.434795667884378
12.145281268506933
11.863146647937148
11.5882036941662
11.320269090197257
11.05916419181968

```

Problem 3.2

Rather than assuming a constant value for the albedo α , we will represent the ice-albedo feedback by letting α depend on T :

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T + 10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoproterozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

```
# this function computes the increment by which the concentration
# is updated at each step
function box_simulate_timestep(T, xi, x, S, C, A, B)
    if T <= -10
        x = xi
    elseif -10 <= T <= 10
        x = xi + (x-xi)*((T+10)/20)
    elseif T >= 10
        x = x
    end

    return ((1-x) * S)/(4*C) - (A-(B*T))/C
end

# this function loops over the timesteps to simulate
# the concentration series
function climate_simulate(xi, x, S, C, A, B, T, Tf, Δt)
    # initialize T storage
    # for code simplicity we make the array length T+1 so
    # index 1 is T
    steps = Int64(Tf / Δt)
    Ts = zeros(steps + 1)
    Ts[1] = T
    for t = 1:steps
        Ts[t+1] = Ts[t] +
            Δt * box_simulate_timestep(Ts[t], xi, x, S, C, A, B)
    end
    # the first element of T is the initial condition
    return Ts
end

Δt = 1.0 #Time step of 1 year
```

```

Tf = 200 #End-time of 200 years
C = 51 #J/m^2/degrees C
x = 0.3 #Planetary albedo alpha
xi = 0.5
x = 0.3
S = 1272 #W/m^2 #Neoproterozoic value
A = 221.2 #W/m^2
B = -1.3 #W/m^2/degrees C

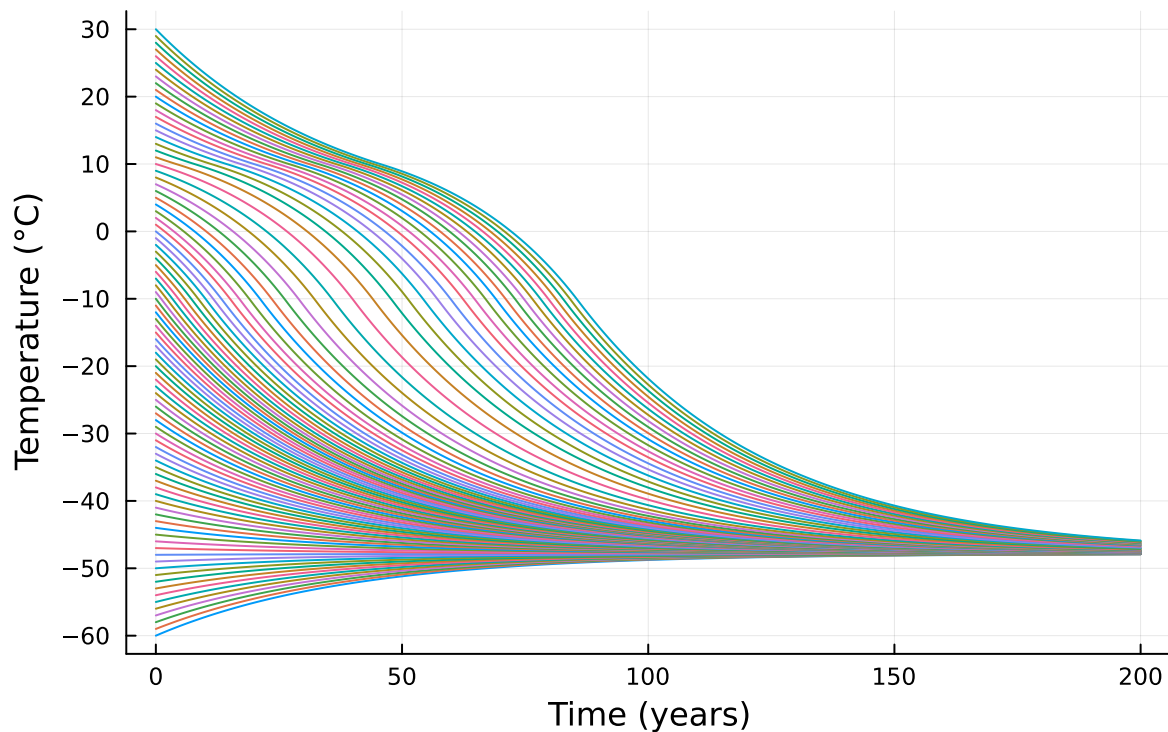
# setting temperature range from -60 C to 30 C
T_values = -60.0:1.0:30.0

Ts = Dict{Float64, Vector{Float64}}{ }()
for T in T_values
    Ts[T] = climate_simulate(xi, x, S, C, A, B, T, Tf, Δt)
end

plot(yticks = -60:10:30)
for (T, T_values) in sort(collect(Ts))
    plot!(0:Δt:Tf, T_values, label="", lw=1)
end
xlabel!("Time (years)")
ylabel!("Temperature (°C)")
title!("Temperature Trajectories, Varying T ")

```

Temperature Trajectories, Varying T_0



From the graph, it is visible that there is 1 equilibrium at approximately -47 degrees Celsius. This equilibrium appears to be relatively stable for temperature values that are at or less than 0 degrees Celsius, but when the temperature is positive, noticeable fluctuations in the slope begin to appear over time in contrast to a steady decline, resulting in an unstable equilibrium for these values.

Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C?

```
# this function computes the increment by which the concentration
# is updated at each step
function box_simulate_timestep(T, xi, x, S, C, A, B)
    if T <= -10
        x = xi
    elseif -10 <= T <= 10
        x = xi + (x - xi) * ((T + 10) / 20)
```

```

elseif T >= 10
    x = x
end

return ((1-x) * S)/(4*C) - (A-(B*T))/C
end
# this function loops over the timesteps to simulate
# the concentration series
function climate_simulate(xi, x, S, C, A, B, T, Tf, Δt)
    # initialize T storage
    # for code simplicity we make the array length T+1 so
    # index 1 is T
    steps = Int64(Tf / Δt)
    Ts = zeros(steps + 1)
    Ts[1] = T
    for t = 1:steps
        Ts[t+1] = Ts[t] +
            Δt * box_simulate_timestep(Ts[t], xi, x, S, C, A, B)
    end
    # the first element of T is the initial condition
    return Ts
end
Δt = 1.0 #Time step of 1 year
Tf = 200 #End-time of 200 years
C = 51 #J/m^2/degrees C
x = 0.3 #Planetary albedo alpha
xi = 0.5
x = 0.3
S = 1368 #W/m^2 #MODERN value
A = 221.2 #W/m^2
B = -1.3 #W/m^2/degrees C

# setting temperature range from -60 C to 30 C
T_values = -60.0:1.0:30.0

Ts = Dict{Float64, Vector{Float64}}{ }()
for T in T_values
    Ts[T] = climate_simulate(xi, x, S, C, A, B, T, Tf, Δt)
end

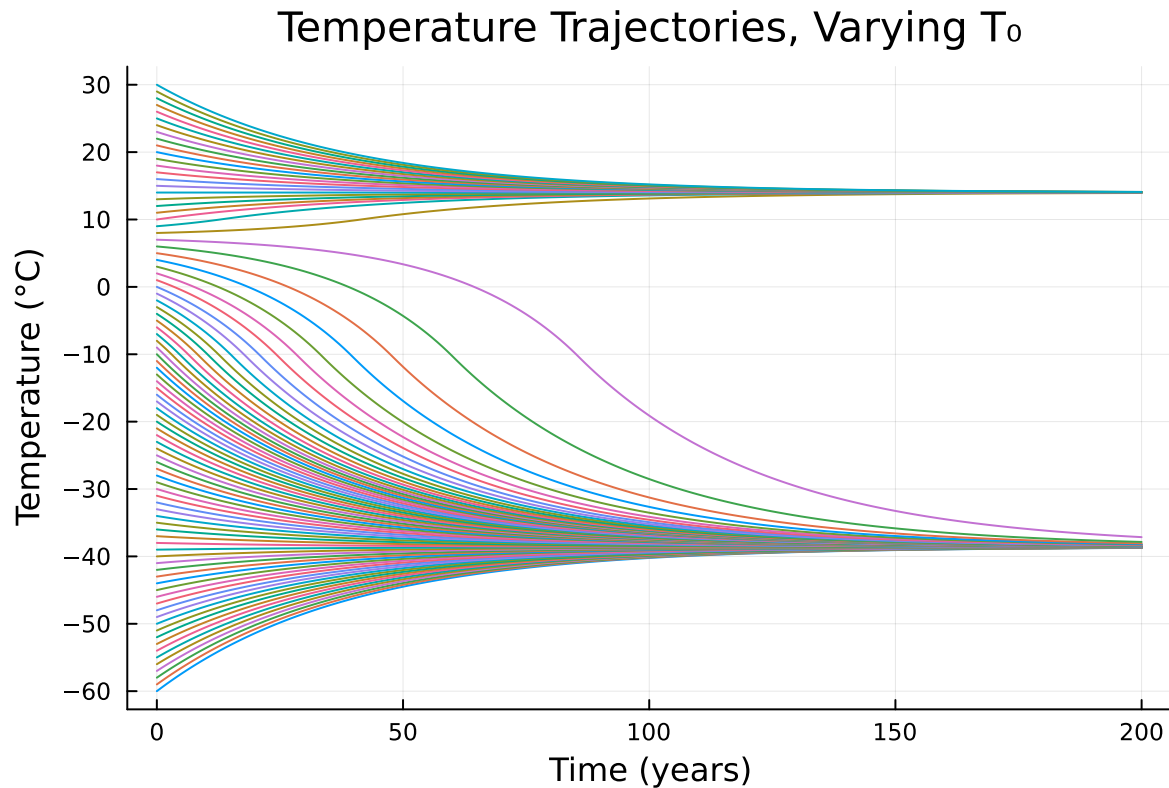
plot(yticks = -60:10:30)
for (T, T_values) in sort(collect(Ts))

```

```

    plot!(0:Δt:Tf, T_values, label="", lw=1)
end
xlabel!("Time (years)")
ylabel!("Temperature (°C)")
title!("Temperature Trajectories, Varying T ")

```



After copying and pasting the same code that was used in 3.2 and only increasing the value of S from 1272 to 1368 W/m^2 , the graph reaches a second equilibrium at approximately $T = 14$ degrees Celsius, given that the initial temperature was at least 8 degrees Celsius. As such, it seems that small increases in temperature over time due to increased solar radiation being received could plausibly have led to an amplifying feedback loop that facilitated the transition from “Snowball Earth” to a pre-industrial planetary temperature of 14 degrees C.

References

List any external references consulted, including classmates.

HW 1 Code * Generating graph for #1

Lecture 6 slides * 3.1 Forward Euler code

<https://discourse.julialang.org/t/how-to-work-with-intervals-values-in-julia/6433/5> * 3.2

Lecture 5 slides * 3.2